

Ocenjevanje natančnosti

$$\begin{aligned}u &= f(x_1, x_2, ..., x_n) = f(\vec{x}), \; \Delta \vec{x}^T = (\Delta x_1, ..., \Delta x_n)^T \\ \Delta u &= \Delta \vec{x}^T \left(\frac{\partial f}{\partial \vec{x}} \right) |_{\vec{x}} \\ (\partial f / \partial \vec{x}) &= (\partial f / \partial x_1, ..., \partial f / \partial x_n)^T \\ \Delta u / u &= (1 / f(x)) \Delta \vec{x}^T \left(\frac{\partial f}{\partial \vec{x}} \right) |_{\vec{x}} \\ \sigma_u^2 &= \sum (\partial f / \partial x_i)^2 \sigma_i^2 + 2 \sum_{i < j} (\partial f / \partial x_i) (\partial f / \partial x_j) \rho_{ij} \sigma_i \sigma_j\end{aligned}$$

Optimalno filtriranje

$$\begin{aligned}\sigma &...nedoločenost(u = u_0 \pm \sigma) \\ \sigma^2 &...varianca\end{aligned}$$

Za izmerek z_i količine x privzamemo $z_i = x + r_i$, $r_i \sim N(0, \sigma_i)$. Predpostavka je smiselna zaradi centralnega limitnega izreka.

$$\frac{d f_{\text{Gauss}}}{d x} = \frac{1}{\sqrt{2 \pi} \sigma} e^{-\frac{(x-\bar{x})^2}{2 \sigma^2}}$$

Združevanje meritev

Več neodvisnih meritev z enako disperzijo

Imamo meritve $\{(z_i, \sigma_i)\}_{i=1}^N$.

$$\begin{aligned}\bar{z} &= \frac{1}{N} \sum_{i=1}^N z_i \\ \sigma_{\bar{z}} &= \frac{\sigma}{\sqrt{N}} \\ \bar{z} &\sim N(x, \sigma_{\bar{z}})\end{aligned}$$

Dve neodvisnih meritvi z različno disperzijo

Imamo meritvi $(\bar{z}_1, \sigma_1 = \frac{\sigma}{\sqrt{N}})$, $(\bar{z}_2, \sigma_2 = \frac{\sigma}{\sqrt{M}})$.

$$\begin{aligned}\bar{z}_3 &= \frac{N}{N+M} \bar{z}_1 + \frac{M}{N+M} \bar{z}_2 = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} \bar{z}_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \bar{z}_2 = \\ &= \bar{z}_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} (\bar{z}_2 - \bar{z}_1) = \sigma_3^2 \sum \frac{\bar{z}_i}{\sigma_i^2} \\ \sigma_3^{-2} &= \sigma_1^{-2} + \sigma_2^{-2}\end{aligned}$$

Dve odvisni meritvi

$$\begin{aligned}\sigma_{xy} &= \langle (x - \bar{x})(y - \bar{y}) \rangle = \langle xy \rangle - \bar{x} \cdot \bar{y} \quad \text{(kovarianca)} \\ \rho_{xy} &= \frac{\sigma_{xy}}{\sigma_x \sigma_y} \quad \text{(korelacijski koeficient)}\end{aligned}$$

Imamo meritvi $(\bar{z}_1, \sigma_1)$ in $(\bar{z}_2, \sigma_2)$ s korelacijo ρ .

$$\bar{z}_3 = \bar{z}_1 + \frac{\sigma_1^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} (\bar{z}_2 - \bar{z}_1) = \bar{z}_1 + K (\bar{z}_2 - \bar{z}_1)$$

$$K = \frac{\sigma_1^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \quad \text{(ojačanje)}$$

$$\sigma_3^2 = (1 - \rho^2) \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} - \frac{2\rho}{\sigma_1 \sigma_2} \right)^{-1}$$

Povprečenje (ni nujno optimalno)

Imamo N meritev $z_i \sim N(x, \sigma_i)$.

$$\bar{z} = \frac{1}{N} \sum_i z_i$$

$$\sigma_{\bar{z}}^2 = \frac{1}{N^2} \left(\sum_i \sigma_i^2 + 2 \sum_{i < j} \sigma_{ij} \right)$$

Širjenje napak

Imamo $(\bar{x}, \sigma_{\bar{x}})$ in $(\bar{y}, \sigma_{\bar{y}})$.

$$\bar{u} \approx f(\bar{x}, \bar{y})$$

$$\sigma_{\bar{u}}^2 \approx \left(\frac{\partial f}{\partial x}(\bar{x}, \bar{y}) \right)^2 \sigma_{\bar{x}}^2 + \left(\frac{\partial f}{\partial y}(\bar{x}, \bar{y}) \right)^2 \sigma_{\bar{y}}^2 + 2 \frac{\partial f}{\partial x}(\bar{x}, \bar{y}) \frac{\partial f}{\partial y}(\bar{x}, \bar{y}) \sigma_{\bar{x}\bar{y}}$$

Merjenje skalarne konstante

Merimo konstanto x z izmerki $z_i = x + r_i$, $r_i \sim N(0, \sigma)$.

Predpostavimo neodvisnost meritev $\langle r_i r_j \rangle = \sigma^2 \delta_{ij}$.

Predstavljamo si merjenje v diskretnih razmikih T .

$$\begin{aligned}\hat{x}_n &= \frac{1}{n} \sum_{i=1}^n z_i \quad \text{(najboljša ocena za } x) \\ \sigma_n^2 &= \frac{\sigma^2}{n}\end{aligned}$$

Diskretno

$$\hat{x}_{n+1} = \hat{x}_n + \frac{\sigma_n^2 + 1}{\sigma^2} (z_{n+1} - \hat{x}_n)$$

$$\sigma_{n+1}^{-2} = \sigma_n^{-2} + \sigma^{-2}$$

Zvezno

$$R = \lim_{T \rightarrow 0} \sigma^2 T > 0$$

$$\langle r(t), r(t') \rangle = R \delta(t - t') \quad \text{(beli šum)}$$

$$\dot{\hat{x}} = \frac{\sigma_{\hat{x}}^2(t)}{R} (z(t) - \hat{x}(t))$$

$$\dot{\sigma}_{\hat{x}}^2(t) = -\frac{\sigma_{\hat{x}}^4(t)}{R}$$

Uteženo povprečevanje

Zbrali smo vse meritve (h_i, σ_i^2)

$$\hat{h} = (\sum h_i / \sigma_i^2) / (\sum \sigma_i^{-2})$$

$$\hat{\sigma}^{-2} = \sum \sigma_i^{-2}$$

Merjenje skalarne spremenljivke

Merimo $x = x(t)$ z izmerki $z_i = x + r_i$, $r_i \sim N(0, \sigma)$ v časovnih razmikih T .

Znana dinamika: $\dot{x}(t) = A(t)x(t) + c(t)$

Diskretno

Dinamika: $x_{n+1} = \phi_n x_n + c_n + \Gamma_n w_n$

$$\phi_n = 1 + A(nT)T$$

$$c_n = c(nT)T$$

$$w_n \sim N(0, Q_n), \quad \langle w_n w_{n'} \rangle = Q_n \delta_{nn'} \quad \text{(dinamični šum)}$$

$$\bar{x}_{n+1} = \phi_n \hat{x}_n + c_n \quad \text{(napoved)}$$

$$M_{n+1} = \bar{\sigma}_{n+1}^2 = \phi_n^2 P_n + \Gamma_n^2 Q_n$$

$$(z_{n+1}, R_{n+1}) \quad \text{(meritev)}$$

$$P_{n+1} = \hat{\sigma}_{n+1}^2 = \frac{M_{n+1} R_{n+1}}{M_{n+1} + R_{n+1}} = M_{n+1} - \frac{M_{n+1}^2}{M_{n+1} + R_{n+1}}$$

$$K_{n+1} = \frac{P_{n+1}}{\sigma^2} \quad \text{(ojačevalni faktor)}$$

$$\hat{x}_{n+1} = \bar{x}_{n+1} + K_{n+1} (z_{n+1} - \bar{x}_{n+1}) \quad \text{(nova ocena)}$$

Zvezno

Dinamika: $\dot{x}(t) = A(t)x(t) + c(t) + \Gamma(t)w(t)$

$$R(t) = \lim_{T \rightarrow 0} \sigma^2 T$$

$$P(t) = \lim_{T \rightarrow 0} P_{n+1}$$

$$Q(t) = \lim_{T \rightarrow 0} T Q_n$$

$$\langle w(t)w(t') \rangle = \delta(t - t')Q(t)$$

$$\Gamma(t) = \lim_{T \rightarrow 0} \frac{\Gamma_n}{T}$$

$$\dot{\hat{x}}(t) = A(t)\hat{x}(t) + c(t) + \frac{P(t)}{R(t)} (z(t) - \hat{x}(t))$$

$$\dot{P}(t) = 2A(t)P(t) - \frac{P^2(t)}{R(t)} + \Gamma^2(t)Q(t)$$

Merjenje vektorske spremenljivke

Merimo $\mathbf{x}(t) = (x_1(t), x_2(t), \dots, x_n(t))$ z izmerki $\mathbf{z} = H\mathbf{x} + \mathbf{r}$, $\mathbf{r} \sim N(0, R)$, $R = \langle \mathbf{r}\mathbf{r}^T \rangle$.

Znana dinamika: $\dot{\mathbf{x}}(t) = A(t)\mathbf{x}(t) + \mathbf{c}(t)$

Diskretno

Dinamika: $\mathbf{x}_{n+1} = \phi_n \mathbf{x}_n + \mathbf{c}_n + \Gamma_n \mathbf{w}_n$

$$\mathbf{w}_n \sim N(0, Q_n), \quad \langle \mathbf{w}_n \mathbf{w}_n^T \rangle = \delta_{nn'} Q_n \quad \text{(dinamični šum)}$$

$$\bar{\mathbf{x}}_{n+1} = \phi_n \hat{\mathbf{x}}_n + \mathbf{c}_n$$

$$M_{n+1} = \langle (\bar{\mathbf{x}} - \mathbf{x})(\bar{\mathbf{x}} - \mathbf{x})^T \rangle = \phi_n P_n \phi_n^T + \Gamma_n Q_n \Gamma_n^T$$

$$P_{n+1} = \langle (\hat{\mathbf{x}} - \mathbf{x})(\hat{\mathbf{x}} - \mathbf{x})^T \rangle =$$

$$= M_{n+1} - M_{n+1} H^T (H M_{n+1} H^T + R_{n+1})^{-1} H M_{n+1}$$

$$K_{n+1} = P_{n+1} H^T R_{n+1}^{-1}$$

$$\hat{\mathbf{x}}_{n+1} = \bar{\mathbf{x}}_{n+1} + K_{n+1} (\mathbf{z}_{n+1} - H \bar{\mathbf{x}}_{n+1})$$

Zvezno

Dinamika: $\dot{\mathbf{x}}(t) = A(t)\mathbf{x}(t) + \mathbf{c}(t) + \Gamma \mathbf{w}(t)$

Oznake v limiti $T \rightarrow 0$ so analogne oznakam pri merjenju skalarne spremenljivke.

$$\dot{\hat{\mathbf{x}}}(t) = A(t)\hat{\mathbf{x}}(t) + \mathbf{c}(t) + K(t)(\mathbf{z}(t) - H\hat{\mathbf{x}}(t))$$

$$K(t) = P(t)H^T R^{-1}(t)$$

$$\dot{P}(t) = A(t)P(t) + P(t)A^T(t) + \Gamma Q(t)\Gamma^T - P(t)H^T R^{-1}(t)HP(t)$$

$$\text{Velja } (AP)^T = PA^T$$

Normalna porazdelitvena funkcija

$$z \sim \frac{dP}{dz} = \frac{1}{\sqrt{2\pi}\sigma} \exp(-(z - \mu)^2 / 2\sigma^2) \rightarrow P(z \leq x) =$$

$$\tilde{F}(x, \mu, \sigma^2) \quad \text{(kumulativna funkcija)}$$

vedemo novo spremenljivko $u = (z - \mu) / \sigma$,

$u \sim 1 / \sqrt{2\pi} \exp(-u^2 / 2) = \mathcal{N}(0, 1)$ (standardizirano normalna porazdelitev)

$$\rightarrow F(x) = \int_{-\infty}^x (dP/du) du \quad \text{(tabelirano)}$$

Lastnosti $F(x) : F(-\infty) = 0$, $F(\infty) = 1$, $F(0) = 1/2$,

$$F(-x) = 1 - F(x)$$

$$z \sim \mathcal{N}(\mu, \sigma^2), \; P(\alpha < z < \beta) = P(z < \beta) - P(z < \alpha) =$$

$$F((\beta - \mu) / \sigma) - F((\alpha - \mu) / \sigma)$$

$$P(\mu - \sigma < z < \mu + \sigma) = 68\%$$

$$P(\mu - 2\sigma < z < \mu + 2\sigma) = 95\%$$

$$P(\mu - 3\sigma < z < \mu + 3\sigma) = 99.7\%$$

$$P(\mu - n\sigma < z < \mu + n\sigma) = 2F(n) - 1$$

Enačbe

Toni

$\Delta t \cdot \Delta \nu \approx 1$, Δt čas trajanja; $\Delta \nu$ nedoločenost frekvence

Oktava: $12 \text{ poltonov}, \nu_{i+1} / \nu_i = x = \text{konst.}$

Senzorji

Senzor kot univerzalni merilni sistem

Red senzorja je red diferencialne enačbe, ki povezuje vhodne $(z(t))$ in izhodne $(\hat{x}(t))$ količine.

Senzor n -tega reda je optimalen sistem za sledenje spremenljivki z dinamiko $\frac{d^n x}{dt^n} = w(t)$, kjer je $w(t)$ beli šum.

Pri Kalmanovem filtru senzorja tipično vzamemo konstanten ojačevalni faktor $K \approx \lim_{t \rightarrow \infty} K(t) = K_{\infty}$.

Za izračun odziva senzorja($x(t)$) na nek signal $z(t)$ uporabimo formulo $x(s) = H(s)z(s)$.

Senzor 1. reda

$$\tau \dot{\hat{x}} + \hat{x} = z$$

$$\tau = \sqrt{\frac{R}{Q}}$$

Senzor 2. reda

$$\ddot{x} + 2\zeta\omega\dot{x} + \omega^2\hat{x} = \omega^2 z(+2\zeta\omega\dot{z})$$

$$\omega^4 = \frac{Q}{R}$$

$$\zeta = \frac{1}{\sqrt{2}} \quad (\text{optimalna vrednost koeficienta dušenja})$$

Prenosna funkcija senzorja

$$\hat{x}(s) = \mathcal{L}(x(t)) = \int_0^\infty e^{-st} f(t) dt$$

$$z(s) = \mathcal{L}(z(t))$$

$$H(s) = \frac{\hat{x}(s)}{z(s)} = \mathcal{L}(G(t)) \quad (\text{prenosna funkcija})$$

$$\text{Senzor 1. reda: } H(s) = \frac{1}{1+\tau s}$$

$$\text{Senzor 2. reda: } H(s) = \frac{\omega^2(+2\zeta\omega s)}{s^2+2\zeta\omega s+\omega^2}$$

Vsako prenosno funkcijo, ki sledi iz LDE s konstantnimi koeficienti, lahko zapišemo kot produkt prenosnih funkcij 1. in 2. reda.

Laplace: Bronštejn 903

Laplace za reševanje sistema dif. enačb

Dinamiko opišemo z matriko in vektorji

Dobimo enačbo za $\dot{P}$

Člene p_{ij} zapišemo v s domeni in rešimo algebraične enačbe

Transformiramo $p_{ij}(s)$ v časovno domeno

Periodični vhodni signal

$$z(t) = z_0 e^{i\omega t} \implies x(t) = x_0 e^{i\omega t + i\delta}$$

$$s = i\omega$$

$$\frac{x_0}{z_0} = |H(i\omega)|$$

$$\tan \delta = \frac{\text{Im } H(i\omega)}{\text{Re } H(i\omega)}$$

Bodejev diagram je sestavljen iz grafa $20 \log |H|$ in δ v odvisnosti od $\log \omega\tau$.

Termični šum na uporniku

$$\frac{d\langle U^2 \rangle}{d\omega} = \frac{2R}{\pi} \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_B T}} - 1}$$

Elektronika

Predpostavimo, da tok v veji z U_{out} ne teče.

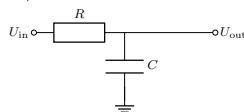
Bodejev diagram narišemo tako, da pogledamo limite $\omega\tau \ll 1$, $\omega\tau = 1$ in $\omega\tau \gg 1$. V teh limitah izračunamo $10 \log |H|^2$.

$$Z_R = R, Z_c = 1/(i\omega C), Z_L = i\omega L.$$

Pasivna vezja

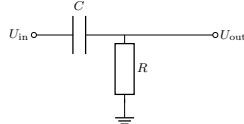
Nizkofrekvenčni prepustni (low-pass) filter

$$H = \frac{U_{\text{out}}}{U_{\text{in}}} = \frac{1}{1+i\omega\tau}, \quad \tau = RC$$



Visokofrekvenčni prepustni (high-pass) filter

$$H = \frac{i\omega\tau}{1+i\omega\tau}, \quad \tau = RC$$

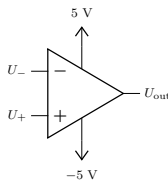


Aktivna vezja

Operacijski ojačevalec (OPAMP)

$$U_{\text{out}} = A(s)(U_+ - U_-)$$

$$\text{Idealni OPAMP: } A_s \rightarrow \infty, I_+ = I_- = 0$$



Negativna povratna zanka

$$U_{\text{out}}(t) = A(U_{\text{in}} - F U_{\text{out}})$$

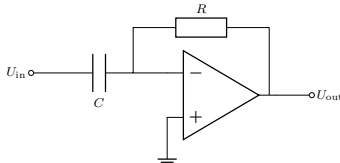
$$H = 1/(A^{-1} + F) \approx 1/F(s)$$

Izgleda kot diferenciator brez kondenzatorja, namesto R imamo škatlo s prenosno funkcijo $F(s)$.

Diferenciator

$$U_{\text{out}}(t) = \tau \dot{U}_{\text{in}}(t)$$

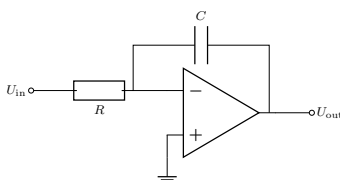
$$H = i\omega\tau, \quad \tau = RC$$



Integrator

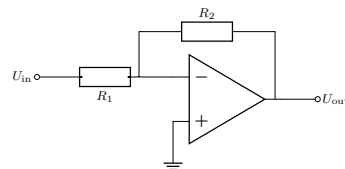
$$U_{\text{out}}(t) = -\frac{1}{\tau} \int_0^t U_{\text{in}}(t') dt'$$

$$H = -\frac{1}{i\omega\tau}, \quad \tau = RC$$



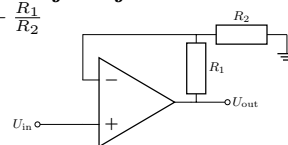
Invertirajoči ojačevalnik

$$H = -\frac{R_2}{R_1}$$



Neinvertirajoči ojačevalnik

$$H = 1 + \frac{R_1}{R_2}$$



Resonančna vezja

$$\text{Izračunamo } |H(i\omega)|^2 = 1/2 \Rightarrow \omega_{1,2}$$

$$\text{Atenuacija } \Delta\omega_{-3\text{ dB}} = \omega_2 - \omega_1$$

Dobrota $Q = \omega_0/\Delta\omega_{-3\text{ dB}}$; če $Q \gg 1$, je resonanca ostra

Pasovni prepustni (band-pass) filter

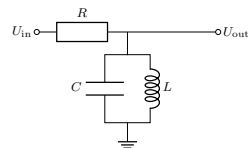
$$H = \frac{s\omega_c}{s^2 + s\omega_c + \omega_0^2}$$

$$\omega_0^2 = \frac{1}{LC}, \quad \omega_c = \frac{1}{RC}$$

Blizu resonance $s \approx i\omega_0$:

$$\Delta\omega_{-3\text{ dB}} = \omega_c$$

$$Q = \frac{\omega_0}{\Delta\omega_{-3\text{ dB}}} = \frac{\omega_0}{\omega_c}$$



Pasovni neprepustni (band-reject) filter

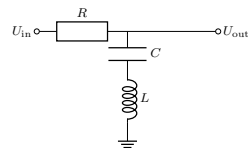
$$H = \frac{s^2 + \omega_0^2}{s^2 + s\omega_c + \omega_0^2}$$

$$\omega_0^2 = \frac{1}{LC}, \quad \omega_c = \frac{R}{L}$$

Blizu resonance $s \approx i\omega_0$:

$$\Delta\omega_{-3\text{ dB}} = \omega_c$$

$$Q = \frac{\omega_0}{\Delta\omega_{-3\text{ dB}}} = \frac{\omega_0}{\omega_c}$$



Statistika

Ocenjevanje parametrov normalne porazdelitve

Imamo vzorec $\{z_i\}_{i=1}^n, z_i \sim N(a, \sigma)$.

$$\bar{z} = \frac{1}{n} \sum_i z_i$$

Poznamo a : $\sigma^2 = \frac{1}{n} \sum_i (z_i - a)^2$

Ne poznamo a : $s^2 = \frac{1}{n-1} \sum_i (z_i - \bar{z})^2$

Studentova porazdelitev:

$$S(k) = \frac{dP}{dT}(k) = \frac{1}{\sqrt{k}B(\frac{k}{2}, \frac{1}{2})} \left(1 + \frac{T^2}{k}\right)^{-\frac{k+1}{2}}$$

$$T = \frac{\bar{z}-a}{\sqrt{s^2}} \sqrt{n} \sim S(n-1)$$

Simetrična na y os

Določimo $T_{>,<}$ s pomočjo $P(T > , < T_{>,<}) = \alpha/2$; velja

$$T_{<} = -T_{>}$$

V tabelah: $P(|T| > T_{>}) = p$, v prvi vrstici je napisan $\alpha[\%]$, v prvem stolpcu pa $(n-1) = k$.

Velja $a = \bar{z} \pm T_{>} \frac{S}{\sqrt{N}}$ na stopnji tveganja α

χ^2 porazdelitev:

$$\chi^2(k) = \frac{dP}{d\chi^2}(k) = \frac{1}{2^{\frac{k}{2}} \Gamma(\frac{k}{2})} (\chi^2)^{\frac{k-2}{2}} e^{-\frac{\chi^2}{2}}$$

$$\chi^2 = (n-1) \frac{s^2}{\sigma^2} \sim \chi^2(n-1)$$

$\chi^2 > 0$, ni simetrična, vrh pri $k-3$

Dvostransko testiranje:

$$P(\chi^2 > \chi^2_{>}) = \alpha/2$$

$$P(\chi^2 < \chi^2_{<}) = \alpha/2 \rightarrow P(\chi^2 > \chi^2_{<}) = 1 - \alpha/2$$

Enostranski test(če nočemo odrezati opcij, kjer je ujemanje zelo dobro):

$$P(\chi^2 > \chi^2_{>}) = \alpha$$

V tabelah: $P(\chi^2 > \chi^2_{>}) = p$, v prvi vrstici je napisan $\alpha[\%]$, v prvem stolpcu pa $(n-1) = k$.

Velja $\sigma^2_{<,>} = (n-1) \frac{s^2}{\chi^2_{>,<}}$ (PAZI ker sta znaka $i, \hat{i}$ zamenjana)

na stopnji tveganja α .

$$< \chi^2_{>} > = n - 1 = k.$$

Splošnejše definicije:

$$X_i \sim N(0, 1) \implies \chi^2 = \sum_{i=1}^n X_i^2 \sim \chi^2(n)$$

$$X \sim N(0, 1), Y \sim \chi^2(n) \implies T = \frac{X}{\sqrt{Y/n}} \sim S(n)$$

Primerjava dveh vzorcev

Imamo vzorca $\{x_i\}_{i=1}^{n_x}$ in $\{y_i\}_{i=1}^{n_y}, x_i \sim N(a_x, \sigma),$

$y_i \sim N(a_y, \sigma)$.

$$T = \frac{(\bar{x}-\bar{y})-(a_x-a_y)}{\sqrt{\frac{1}{n_x} + \frac{1}{n_y}}} \frac{\sqrt{n_x+n_y-2}}{\sqrt{s_x^2(n_x-1)+s_y^2(n_y-1)}} \sim S(n_x+n_y-2)$$

Imamo vzorca $\{x_i\}_{i=1}^{n_x}$ in $\{y_i\}_{i=1}^{n_y}, x_i \sim N(a_x, \sigma_x),$

$y_i \sim N(a_y, \sigma_y)$.

$$F = \frac{s_x^2/\sigma_x^2}{s_y^2/\sigma_y^2} \sim F(n_x-1, n_y-1)$$

Oblikovni testi

Pearsonov test χ^2

Imamo izmerke $\{z_i\}_{i=1}^n$ razdeljene v ρ razredov (intervalov z mejami $\xi_0, \dots, \xi_\rho$). Število izmerkov v k -tem intervalu je N_k .

Testiramo porazdelitveno funkcijo $\frac{dP}{d\xi}$.

Uporabimo enostranski test.

$$P_k = \int_{\xi_{k-1}}^{\xi_k} \frac{dP}{d\xi} d\xi$$

$$\chi^2 = \sum_{k=1}^{\rho} \frac{(N_k - nP_k)^2}{nP_k} \sim \chi^2(\rho-1) \quad (\text{če } nP_k \gtrsim 5 \forall k)$$

Fisherjev izrek

Imamo izmerke $\{z_i\}_{i=1}^n$ razdeljene v ρ razredov (intervalov z mejami $\xi_0, \dots, \xi_\rho$). Število izmerkov v k -tem intervalu je N_k .

Testiramo porazdelitveno funkcijo z neznanimi parametri

$$\frac{dP}{d\xi}(q_1, \dots, q_m).$$

$$P_k(q_1, \dots, q_m) = \int_{\xi_{k-1}}^{\xi_k} \frac{dP}{d\xi}(q_1, \dots, q_m) d\xi$$

$$L^* = \prod_{k=1}^{\rho} [P_k(q_1, \dots, q_m)]^{N_k}$$

Optimalne vrednosti q_i^* da sistem $\frac{\partial L^*}{\partial q_i} = 0, i = 1, \dots, m$.

$$\chi^2 = \sum_{k=1}^{\rho} \frac{(N_k - nP_k(q_1^*, \dots, q_m^*))^2}{nP_k(q_1^*, \dots, q_m^*)} \sim \chi^2(\rho - m - 1)$$

Poenostavljena verzija (velja za $n \gg m$):

$$L = \prod_{k=1}^n \frac{dP}{d\xi}(z_k; q_1, \dots, q_m)$$

Optimalne vrednosti $\hat{q}_i$ da sistem $\frac{\partial L}{\partial q_i} = 0, i = 1, \dots, m$.

$$\chi^2 = \sum_{k=1}^{\rho} \frac{(N_k - nP_k(\hat{q}_1, \dots, \hat{q}_m))^2}{nP_k(\hat{q}_1, \dots, \hat{q}_m)} \sim \chi^2(\rho - m - 1)$$

Test Kolmogorova

Imamo izmerke $\{z_i\}_{i=1}^n$, testiramo porazdelitveno funkcijo $\frac{dP}{d\xi}$.

$$F(z) = \int_{-\infty}^z \frac{dP}{d\xi} d\xi$$

$$f(z) = \frac{\{\text{števílo izmerkov manjših od } z\}}{n}$$

$$D = \sup |F(z) - f(z)|$$

$$\lim_{n \rightarrow \infty} P(D\sqrt{n} < d) = \sum_{k=-\infty}^{\infty} (-1)^k e^{-2k^2 d^2}$$

Metoda najmanjših kvadratov

Imamo n meritev $z_i = z(t_i)$ in fit funkcije $\{f_i\}_{i=0}^m$. Iščemo uteži $\{x_i\}_{i=0}^m$.

$$\text{Nekorelirane meritve z disperzijo } \sigma \implies R = \sigma^2 I$$

$$\mathbf{z} = \begin{bmatrix} z(t_1) \\ \vdots \\ z(t_m) \end{bmatrix} = H\mathbf{x} = \begin{bmatrix} f_0(t_1) & f_1(t_1) & \cdots & f_m(t_1) \\ \vdots & \vdots & \ddots & \vdots \\ f_0(t_n) & f_1(t_n) & \cdots & f_m(t_n) \end{bmatrix} \begin{bmatrix} x_0 \\ \vdots \\ x_m \end{bmatrix}$$

$$\hat{\mathbf{x}} = (H^T R^{-1} H)^{-1} H^T R^{-1} \mathbf{z} = (H^T H)^{-1} H^T \mathbf{z}$$

$$P = (H^T R^{-1} H)^{-1} = (H^T H)^{-1} \sigma^2$$

$$2J = (\mathbf{z} - H\mathbf{x})^T R^{-1} (\mathbf{z} - H\mathbf{x}) \sim \chi^2(n - m - 1)$$

Linearni model

$z = x_1 t$, imamo n izmerkov $z_i = z(t_i)$

$$\hat{x} = (\sum t_i z_i) / (\sum t_i^2)$$

$$z = x_0 t + x_1 \implies f_0(x) = x, \quad f_1(x) = 1$$

Imamo n izmerkov $z_i = z(t_i)$.

$$\hat{\mathbf{x}} = \frac{1}{n \sum t_i^2 - (\sum t_i)^2} \begin{bmatrix} n \sum t_i z_i - (\sum t_i)(\sum z_i) \\ -(\sum t_i)(\sum t_i z_i) + (\sum z_i)(\sum t_i^2) \end{bmatrix}$$

$$\hat{x}_0 = \frac{\bar{t}z - \bar{t} \cdot \bar{z}}{\bar{t}^2 - \bar{t}^2}$$

$$\hat{x}_1 = \bar{z} - \bar{t} \hat{x}_0$$

Nelinearen model

Npr. $I_i = I_0 \exp(-t_i/\tau)$

Prevedemo na $\ln(I_i/I_0) = -t_i/\tau = x_0 t_i$.